Economic Analysis of Financial Markets(EAFM)

Economics Module

Lectures 3/4/5/6

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Readings

- 5 – Ch 6 Cost theory (Wilkinson)
- 6 – Ch 7 The Nature of Industry (Baye)
- 7 – Ch 8 Market Structure (Wilkinson)
- 8 – Ch 10 Pricing Strategy (Wilkinson)


COSTS

Introduction

- Many costs are more controllable than are factors affecting revenue
- Must be able to distinguish between short and long run
- Short run
  - Concerned with determining optimal level of output from a given plant size
- Long run
  - All inputs variable
  - Key decision is optimal scale
    - Efficiency
- Different definitions of cost

Distinctions of cost

- Explicit
- Implicit
- Historical
- Current
- Incremental
- Sunk
- Private
- Social

Failing to ignore ***Sunk Costs***

- A **sunk cost** is a cost that has already been paid and cannot be recovered.

- Sunk cost is a past expenditure that cannot be recovered.
  - If an expenditure is sunk, it is not an opportunity cost. So we should not consider it for managerial decisions.
  - However, sunk costs appear in financial accounts.

- Once you have paid money and can't get it back, you should ignore that money in any future decisions you make. But people often allow past costs to influence future decisions.

- *Example: NFL teams persist with first-round-pick quarterbacks much longer than later-round picks with similar performance, because they have "paid" more for the first-rounder.*

- Admitting mistakes and moving on is crucial, but people often find that difficult to do.

A blogger who understands ***Sunk Costs***

- In 2000, Arnold Kim began blogging about Apple products. By 2008, Kim's site had become very successful, and he was earning more than $100,000 per year from paid advertising.
- Sounds good, right? The "problem" was, Kim was a medical doctor who had invested over $200,000 in his education.
- What should Kim do? He believed he would ultimately make more as a blogger than as a doctor but committing to blogging full-time would mean "wasting" his education.
- Kim realized his education costs were sunk—unrecoverable regardless of what career choice he made. So he went with what he wanted to do: blogging full time.
- Could you have made the same choice?

Short run cost behaviour

MP_L = marg. prod. of labour
L = labour
q = output

- Fixed costs F
- Variable costs VC
- Unit costs vs total costs C
  - Marginal cost MC
  - Average Variable cost $\overline{VC}$
  - Average fixed cost $\overline{F}$
  - Average total cost $\overline{C}$

The Marginal Cost Curve

❑ $MC = \Delta VC/\Delta q$

✓Marginal cost (MC) is the change in variable cost as output increases by 1 unit.

✓The MC curve is U-shaped because of diminishing marginal returns. DMR

❑ $MC = \Delta VC/\Delta q = w(\Delta L/\Delta q)$

✓In the short run, capital is fixed. So, the change in variable cost as output increases by 1 unit must be the change in the cost of labor.

✓Marginal cost equals the wage times the extra labor necessary to produce 1 more unit of output.

✓In Figure 6.2, to increase Q from 5 to 6 units, MC = \$40 (4 x \$10).

❑ $MC = w/MP_L$

✓Remember $MP_L = \Delta q/\Delta L$. So, $\Delta L/\Delta q$ is just the inverse of MP_L.

✓Marginal cost equals wage divided by marginal product of labor.

✓Marginal product of labor and marginal cost move in opposite directions as output changes.

Short-Run Costs

Common Measures of Cost: Fixed, Variable and Total

Fixed Cost ***F*** **+** **Variable Cost** ***VC*** **=** **Total Cost** ***C = F + VC***

- All firms use the same basic cost measures, and these should be based on input's opportunity costs.

- Fixed Cost (*F*) does not vary with the level of output; includes expenditures on land, office space, production facilities, and other overhead expenses; are often sunk costs, but not always.

- Variable Cost(*VC*) changes as the quantity of output changes; refers to the costs of variable inputs.

  $C = F + VC$

- Total Cost (*C*) is the sum of fixed and variable costs.

- *F* and *VC* should be based on inputs' opportunity costs.

Short-Run Costs

Average Cost (÷ q)

Average Fixed Cost

AFC = F/q

+

Average Variable Cost

AVC = VC/q

=

Average Cost

AC = C/q = AFC + AVC

- Average Fixed Cost (*AFC*) falls as output rises because the fixed F cost is spread over more units.
- Average Variable Cost (*AVC*) or variable VC_u cost per unit of output may either increase or decrease as output rises.
- Average Cost (*AC*) or average total cost may either increase or decrease as output rises.
  - ✓ In Table 6.1, AFC falls with output and AVC eventually rises with output. AC falls until output of 8 units and then rises.

Short-Run Costs

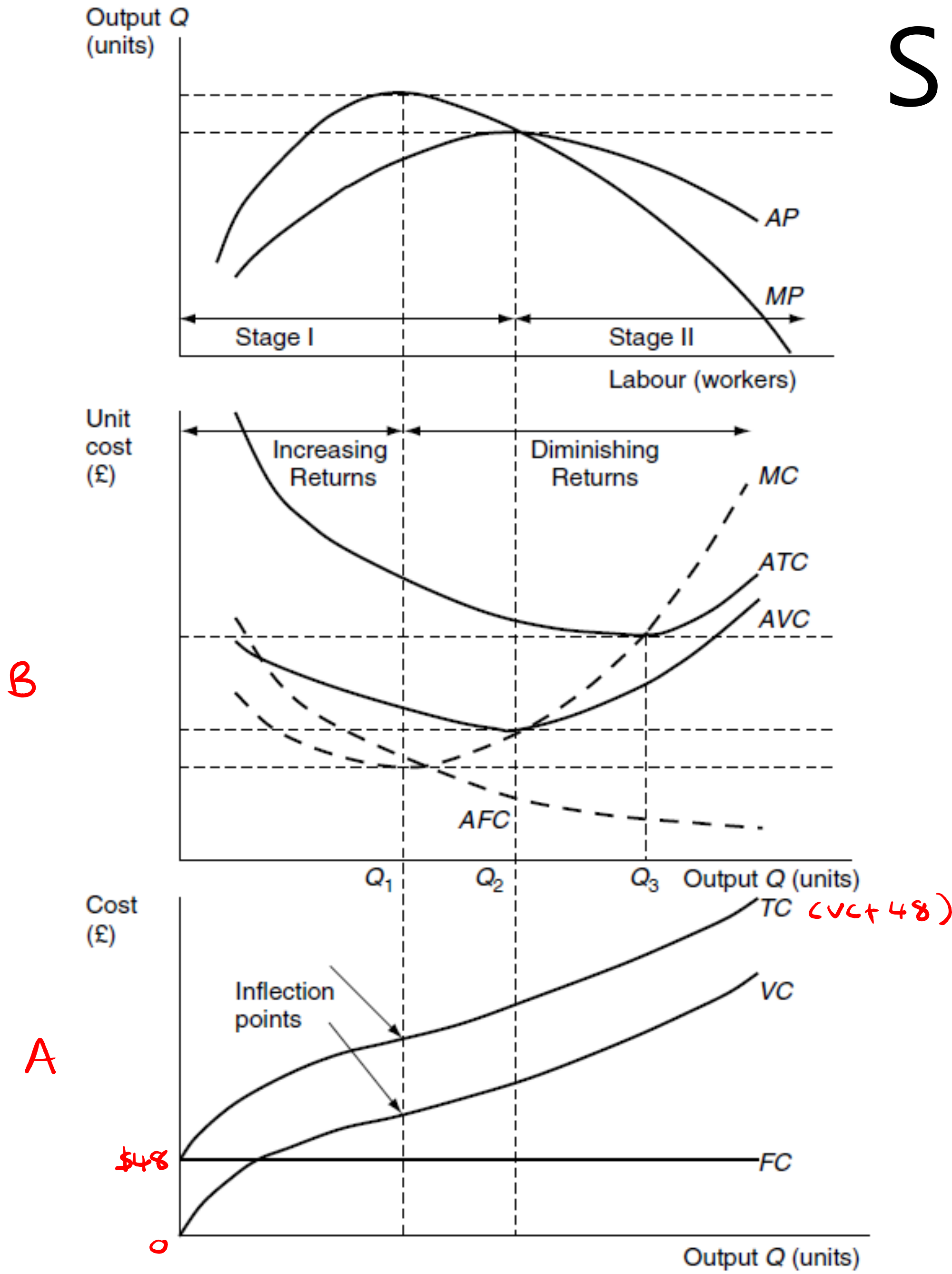


Figure 6.2. Short-run cost and production functions.

- Panel a of Figure 6.1 shows the *VC*, *F*, and *C* curves that correspond to Table 6.1
  - ✓The fixed cost curve, *F,* is a horizontal line at $48.
  - ✓The variable cost curve, *VC,* is zero at zero units of output and rises with output.
  - ✓The total cost curve, *C,* is the vertical sum of the *VC* and *F* curves, so it is $48 higher than the VC curve at every output level. *VC* and *C* curves are parallel.

- Panel b of Figure 6.1 shows the *AFC*, *AVC*, *AC*, and *MC* curves.
  - ✓The marginal cost curve, *MC*, cuts the average variable cost, *AVC*, and average cost, *AC*, curves at their minimums.
  - ✓The height of the *AC* curve at point *a* equals the slope of the line from the origin to the cost curve at *A*.
  - ✓The height of the *AVC* at *b* equals the slope of the line from the origin to the *VC* curve at *B*.
  - ✓The height of the *MC* is the slope of either the *C* or *VC* curve at that quantity

The Average Cost Curves

- *AVC* = VC/*q*
  - ✓Average variable cost is just the variable cost divided by output.

    DMR_L
  - ✓Given the shape of the *VC* is determined by the production function, the diminishing marginal returns to labor also determines the shape of the *AVC* cost curve.
  - ✓So, the VC and AVC curves are both U-shaped.

- *AVC* = VC/*q = wL/q*
  - ✓In the short run, capital is fixed. So the variable cost is *wL*.
  - ✓The average variable cost is cost of labor divided by output.

- $AVC = w/AP_L$

  $AP_L = MP_L$
  - ✓Remember $AP_L = q/L$. So, L/q is just the inverse of AP_L.
  - ✓Average cost equals wage divided by average product of labor.
  - ✓In Figure 6.2, at 6 units of Q, the AVC = $10/0.25 = $40
  - ✓Average product of labor and average variable cost move in opposite directions as output changes.

Production Functions and the Shapes of Short-Run Costs Curves

- The production function determines the shape of a firm's cost curves.

  DMR_L
- In the short run, diminishing marginal returns to labor determine the shape of the production function.
  - ✓the firm increases output by using more labor. However, each extra worker increases output by a smaller amount.

- The production function determines the shape of the variable cost curve and its related curves.
  - ✓As output increases, variable cost increases more than proportionally because of diminishing marginal returns.
  - ✓The production function determines the shape of the marginal cost, average variable cost, and average cost curves.

Long run cost behaviour

- In the long run, the firm adjusts all its inputs so that its cost of production is as low as possible.
- The firm can change its plant size, design, build new machines, and otherwise adjust inputs that were fixed in the short run.
- Fixed costs are avoidable in the long run. They are not sunk costs, as they are in the short run. For instance, the rent a restaurant pays is a fixed cost and this rent can be avoided in the long run if the restaurant does not renew the rental agreement.
- Economies of Scale (EoS)
  - Aspects of increasing scale that lead to falling long-run unit costs
  - Internal EoS
    - Arise from growth of firm itself
    - Four main categories
    a) Technical
    b) Managerial
    c) Marketing
    d) Financial
  - External EoS
    - Arise from growth on the industry
  - Physical/Monetary
  - Level: product, plant and firm

LRAC = Long-run $\bar{C}$

Input Choice

- Technically and Economically Efficient
  - ✓From among the technically efficient combinations of inputs that can be used to produce a given level of output, a firm wants to choose that bundle of inputs with the lowest cost of production, which is the economically efficient combination of inputs.
  - ✓To do so, the firm combines information about technology from the isoquant with information about the cost of production.
- The *LRAC* curve must be U-shaped if a production function has
  - ✓increasing returns to scale at low levels of output,
  - ✓constant returns to scale at intermediate levels of output, and
  - ✓decreasing returns to scale at high levels of output.
- *LRAC* curves can have many different shapes depending whether the production process has economies of scale or diseconomies of scale.
  - ✓A cost function shows economies of scale if the *AC* falls as output expands. Increasing returns to scale in production are sufficient for it.
  - ✓If an increase in output has no effect on AC, the production process has no economies of scale.
  - ✓A cost function shows diseconomies of scale if AC rises when output increases.
  - ✓All these relationships are illustrated in Table 6.3.
- Perfectly competitive firms typically have U-shaped *LRAC* curves.
- Noncompetitive markets may be U-shaped, L-shaped, everywhere downward sloping, everywhere upward sloping or have other shapes.

Long run cost behaviour

- Diseconomies of Scale (DEoS)
  - Aspects of increasing scale that lead to rising long-run unit costs

a) Technical diseconomies
b) Managerial
c) Marketing
d) Transportation

Economies of Scope: The Costs of Producing Multiple Goods

SC

- If a firm produces two or more goods that are linked by a single input, the cost of one good may depend on the output level of another.
    - For example, cattle provide beef and hides.
- A cost function exhibits economies of scope if it is less expensive to produce goods jointly than separately.
    - It is less expensive to produce beef and hides together than separately, so there are economies of scope.
- Economies of Scope: $SC = [C(q_1,0) + C(0,q_2) - C(q_1, q_2)]/C(q_1, q_2)$

$\frac{x+y-xy}{xy}$

- The costs of producing q_1 of the first good, q_2 of the second good and both goods together are $C(q_1, 0)$, $C(0, q_2)$ and $C(q_1, q_2)$.
    - If SC is zero, $C(q_1,0) + C(0,q_2) = C(q_1, q_2)$. There are no economies of scope.
    - If SC is positive, it is cheaper to produce goods jointly. There are economies of scope.
    - If SC is negative, it is cheaper to produce goods separately. There are diseconomies of scope.

Economies of Scale

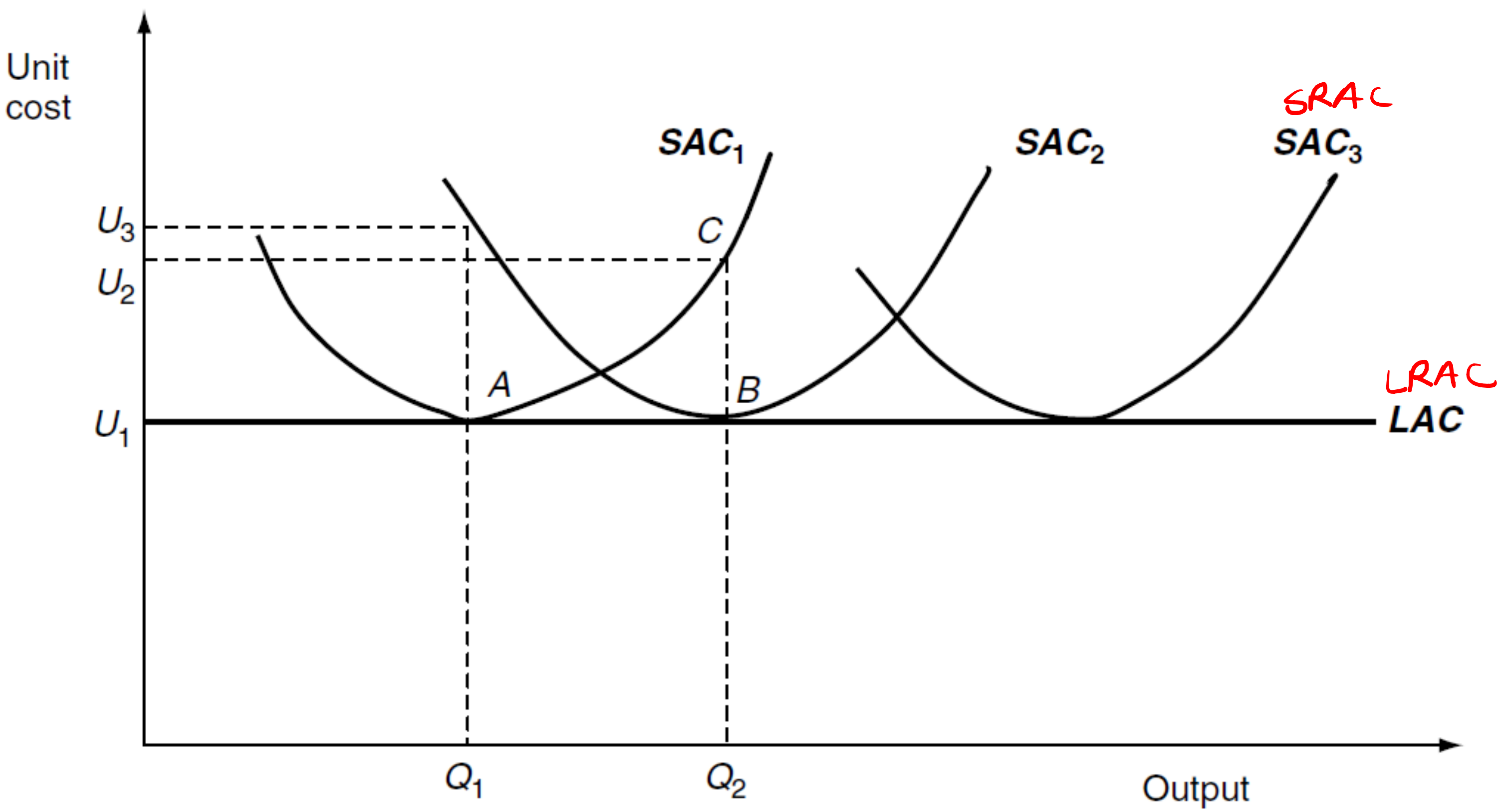


Figure 6.5. Short-run and long-run average cost functions in the absence of EOS and DOS.

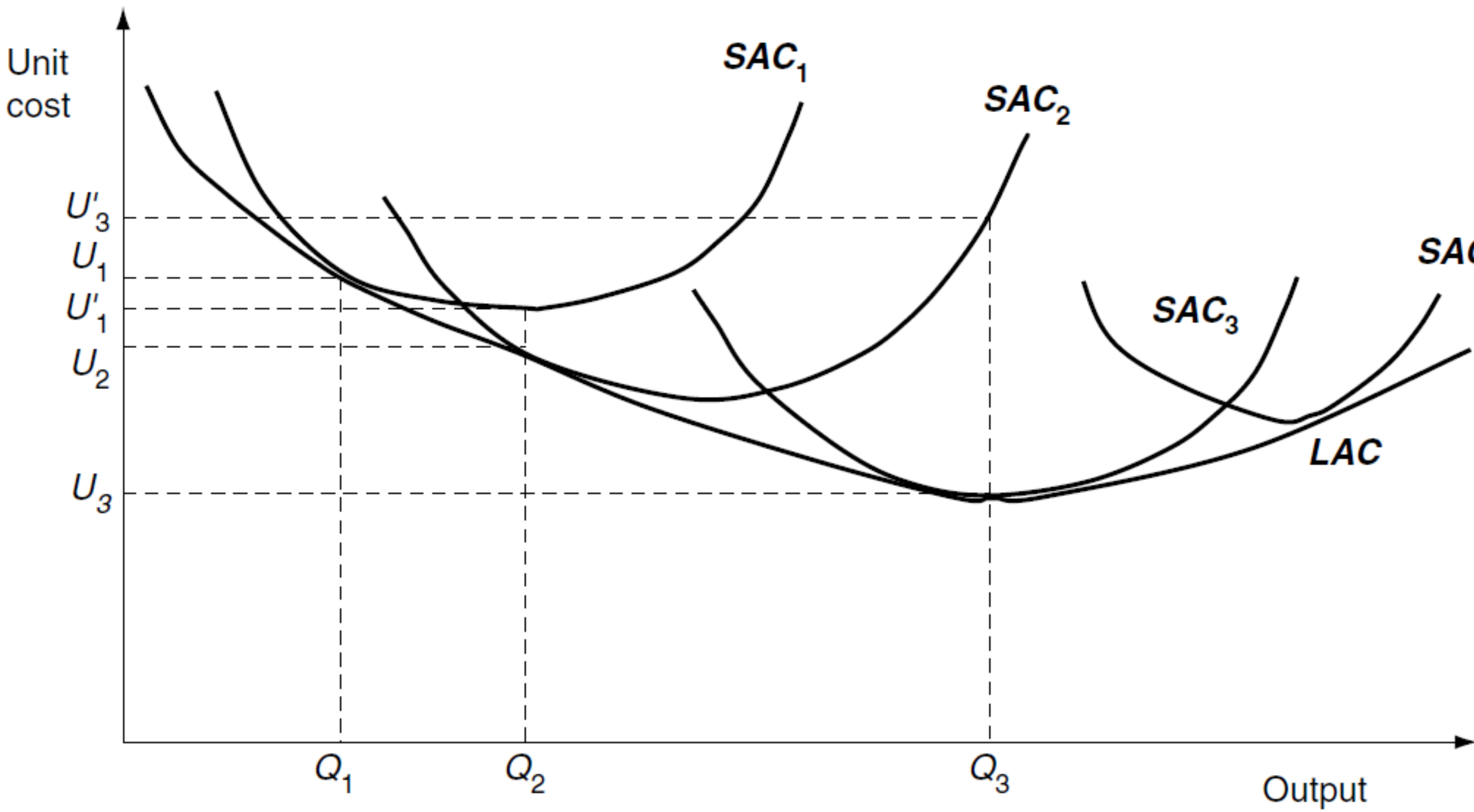


Figure 6.6. Short-run and long-run average cost functions with EOS and DOS.

Relationship btwn short & long run cost curves

a) Optimal scale
b) Minimum efficient scale
c) Multiplant production

The learning curve

- Average cost may fall over time because of increasing IRS returns to scale, technological progress and learning by doing.
- Learning by doing refers to the productive skills and knowledge that workers and managers gain from experience.
  - Workers add speed with practice. Managers learn how to organize production more efficiently, assign tasks based on worker's skills, and reduce inventory costs. Engineers optimize product designs with experimentation.
  - For these and other reasons, the average cost of production tends to fall over time, and the effect is particularly strong with new products.
- The learning curve is the relationship between average costs and cumulative output. The cumulative output is the total number of units of output produced since the product was introduced. Panel a of Figure 6.8 shows the learning curve for Intel central processing unit.
  - If a firm is operating in the economies of scale section of its average cost curve, expanding output lowers its average cost for two reasons: economies of scale, and learning by doing.
  - In panel b of Figure 6.8, economies of scale are seen in each isocost line (AC^1, AC^2 and AC^3), and learning by doing by the distance between those lines.

Learning by Doing

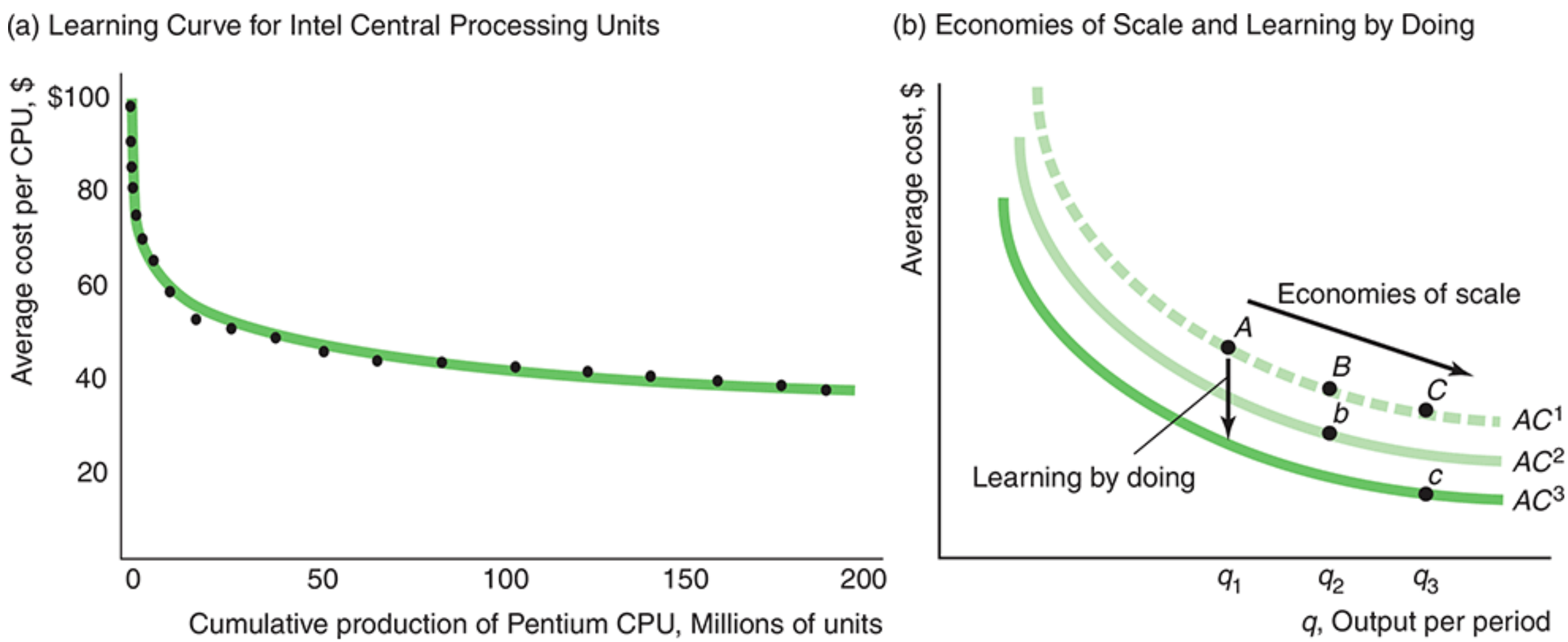


Fig 6.8 (from another textbook)

Short Run Costs

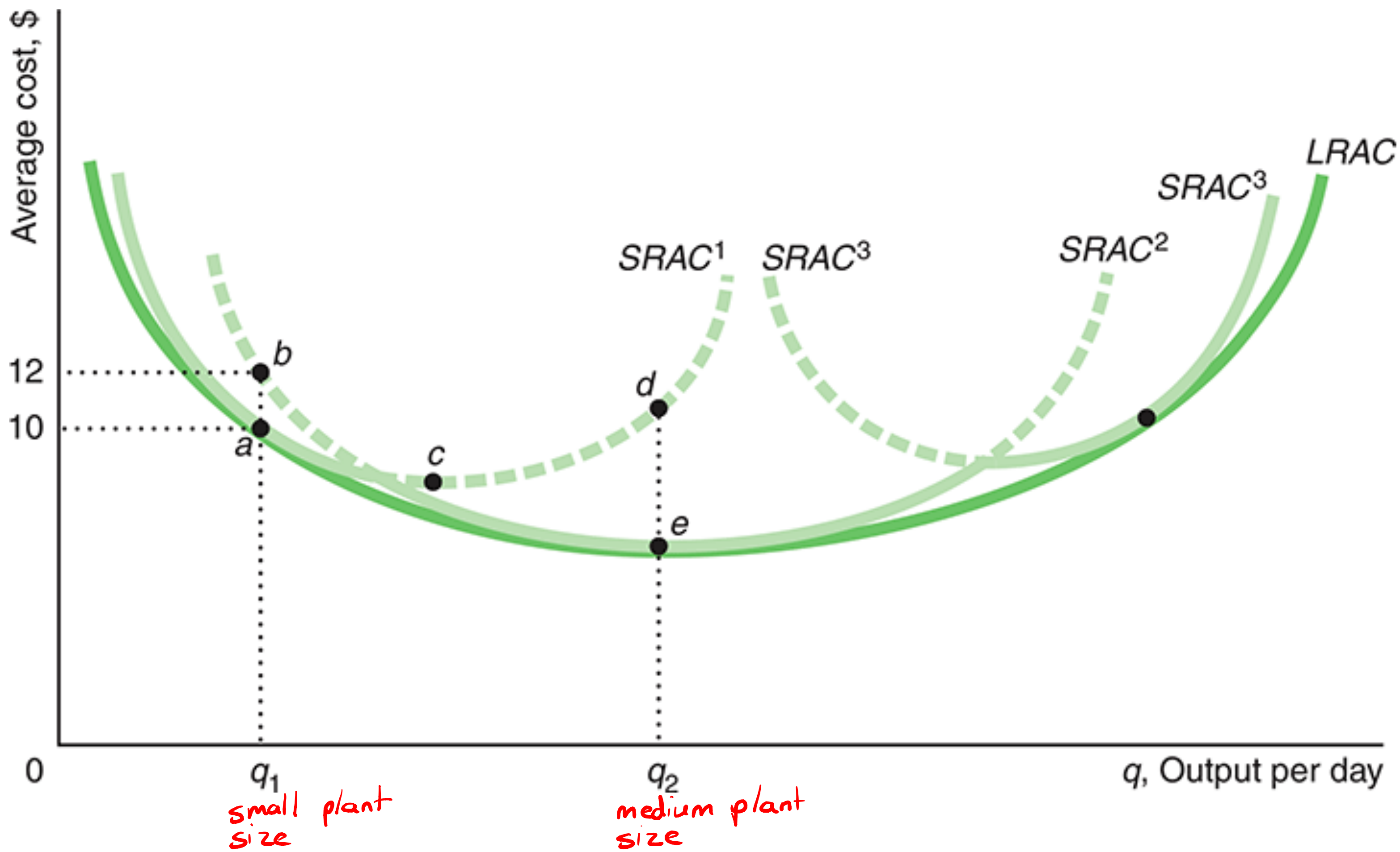
Short-Run Cost Summary

- In the short run, the cost associated with fixed inputs is fixed, while the cost from inputs that can be adjusted is variable.
- Given that input prices are constant, the shapes of the variable cost and the cost-per-unit curves are determined by the production function.
- Where there are diminishing DMR marginal returns, the variable VC cost and cost curves become relatively steep as output increases, so the average cost, $\bar{C}$ average variable $\overline{VC}$ cost, and marginal MC cost curves rise with output.
- The average $\bar{C}$ cost and average variable $\overline{VC}$ cost curves fall when marginal MC cost is below them and rise when marginal cost is above them, so the marginal cost cuts both these average cost curves at their minimum points.

Long Run Costs

LRAC

Figure 6.7 Long-Run Average Cost as the Envelope of Short-Run Average Cost Curves



LRAC as the Envelope of *SRAC* Curves

LRAC ≤ SRAC

- The long-run average cost is always equal to or below the short-run average cost. Any input combination in the short run is also available in the long run. However, changing capital levels to reduce costs are only available in the long run.
- In the long run, the firm chooses the plant size that minimizes its cost of production, so it picks the plant size that has the lowest average cost for each possible output level.
- Graph Analysis
  - ✓ At q_1, in Figure 6.7, the firm opts for the small plant size, whereas at q_2, it uses the medium plant size.
  - ✓ If there are only three possible plant sizes, with short-run average costs $SRAC^1$, $SRAC^2$, and $SRAC^3$, the long-run average cost curve is the solid, scalloped portion of the three short-run curves (envelope curve).
  - ✓ *LRAC* is the smooth and U-shaped long-run average cost curve.

The Behavioral Economics of Shopping

Behavioral economists say that it *does* matter that consumers do not usually make “optimal” consumption choices.

- They believe modeling how people *actually* make decisions is important.

Some important “irrational” consumption behaviors include:

Rules of Thumb

- Making general rules that *often*, but not *always*, produce optimal results
- This can save on decision-making time

Anchoring

- “Irrelevant” information can often influence behavior.
- *Example: posting “limit 10 items per customer” will often induce people to buy 10 items, even they would have bought fewer without the sign*

Making the Connection

J.C. Penney Meets Behavioral Economics

When Ron Johnson became CEO of J.C. Penney, he instituted a new pricing strategy of "everyday low prices", instead of artificially high "regular" prices, and normal "sale" prices.

It turns out that consumers buy *much* more when told an item is on sale, even if the sale price is the same as the "everyday low price".

- This is an example of "anchoring"; the "regular" price acts as an anchor, making people believe they are getting a good deal.

Johnson thought people were smart enough to see through this common department store ploy. But he was wrong, and he paid for his mistake with his job when he was fired after only 17 months.

The Nature of Industry

I. Market Structure
- Measures of Industry Concentration

II. Conduct
- Pricing Behavior
- Integration and Merger Activity

III. Performance
- Dansby-Willig Index
- Structure-Conduct-Performance Paradigm

Industry Analysis

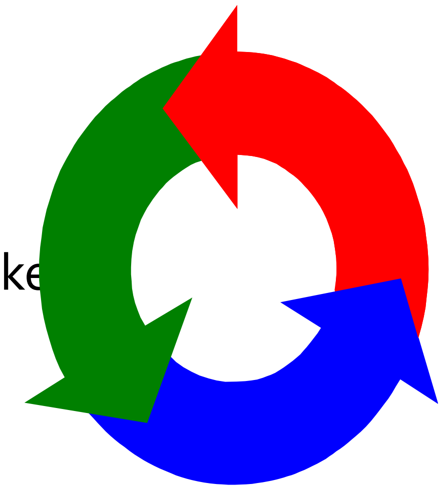
- Market Structure
  - Number of firms, size, etc.
- Conduct
  - Pricing, advertising, R&D, etc.
- Performance
  - Profitability, consumer surplus, social welfare.

The Structure-Conduct-Performance Paradigm

- The Causal View

Market Structure → Conduct → Performance

- The Feedback Critique
  - No one-way causal link.
  - Conduct can affect market structure.
  - Market performance can affect conduct as well as marke structure.



- **Structure-Conduct-Performance** (S-C-P) paradigm: The theory, developed in the 1940’s, says structure affects conduct which affects performance
  - Structure: number of firms in industry, conditions of entry, product differentiation
  - Conduct: pricing strategies, advertising, product development, legal tactics, collusion
  - Performance: maximization of society’s welfare

Market Structure

TABLE 7.1 Market Structure

CHARACTERISTIC	PERFECT COMPETITION	MONOPOLISTIC COMPETITION	OLIGOPOLY	MONOPOLY
Number of firms competing with each other	Large number	Large number	Small number	Single firm
Nature of the product	Undifferentiated	Differentiated	Undifferentiated or differentiated	Unique differentiated product with no close substitutes
Entry into the market	No barriers to entry	Few barriers to entry	Many barriers to entry	Many barriers to entry, often including legal restrictions
Availability of information to market participants	Complete information available	Relatively good information available	Information likely to be protected by patents, copyrights, and trade secrets	Information likely to be protected by patents, copyrights, and trade secrets
Firm's control over price	None	Some	Some, but limited by interdependent behavior	Substantial

TABLE: Measures of Market Power, Selected Manufacturing Industries (1997)

TABLE 8.1 Measures of Market Power, Selected Manufacturing Industries (1997)

Herfindahl – Hirschman Index

INDUSTRY	CR4	CR8	HHI (50 LARGEST COMPANIES)
Meat products	35	48	393
Breakfast cereal	83	94	2,446
Cigarettes	99	N/A	N/A
Sawmills	15	20	87
Book printing	32	45	364
Household refrigerators/freezers	82	97	2,025
Motor vehicles/car bodies	87	94	N/A
Computers	40	68	658

Source: U.S. Department of Commerce, 1997 Economic Census: Concentration Ratios in Manufacturing (2001, Table 2), http://www.census.gov/prod/ec97/m31s-cr.pdf.

Concentration Ratio (CR)

- The concentration ratio is the percentage share of industry sales of the four leading firms in the industry
- Four-firm and eight-firm concentration ratios are more commonly used, concentration ratios can be calculated for any number of top firms. 20 and 50-firm concentration ratios are particularly useful when comparing industries with larger number firms.

Industry Concentration

- **Four-Firm Concentration Ratio**
  - The sum of the market shares of the top four firms in the defined industry. Letting S_i denote sales for firm i and S_T denote total industrv sales

s = sales

$$C_4 = w_1 + w_2 + w_3 + w_4, \; where \; w_1 = \frac{S_{i_{ndividual}}}{S_{T_{otal}}}$$

  - Insensitive to unequal market shares
- **Herfindahl-Hirschman Index (HHI)**
  - Used to understand the level of competition that exists within a market or industry
  - Indication of how the distribution of market share occurs across the companies included in the index
  - The sum of the squared market shares of firms in a given industry, multiplied by 10,000: HHI = 10,000 × Σw_i^2, where $w_i = S_i/S_T$.

w = market shares of firms = $\frac{S_i}{S_T}$

Herfindahl-Hirschman Index (HHI)

- Definition: sum of squared market shares

$$HHI = MS_1^2 + MS_2^2 + MS_3^2 + MS_4^2 \ldots + MS_n^2$$

- HHI can have a theoretical value ranging from close to zero to 10,000.
  - HHI value less than 100, market is highly competitive.
  - HHI value between 100 and 1500, market is not concentrated/unconcentrated.
  - HHI value between 1500 and 2500, market is moderately concentrated.
  - HHI value above 2500, market is highly concentrated.
- If there were a great number of market participants with each company having a market share of almost 0% then the HHI could be close to zero.
  - Perfect Competition?
- Example: a 3 firm industry (30,30,40) has HHI =

$$(30)^2 + (30)^2 + (40)^2 =$$
$$900 + 900 + 1600 = 3400$$

TABLE 9.1 Oligopolistic Industries in the United States

TABLE 9.1 Oligopolistic Industries in the United States

m s

INDUSTRY	NUMBER OF FIRMS	MARKET SHARE (PERCENT)
Carbonated soft drinks	3	80
Beer	3	80
Cigarettes	3	80
Recorded music	4	80
Railroad operations	4	100
Movies	6	85
Razors and razor blades	3	95
Cookies and crackers	2	80
Carpets	2	75
Breakfast cereals	4	80
Light bulbs	2	85
Consumer batteries	3	90

Source: Stephen G. Hannaford, *Market Domination! The Impact of Industry Consolidation on Competition, Innovation, and Consumer Choice* (Westport, CT: Praeger, 2007), 5. Reprinted by permission.

CR

Table 7. Concentration ratio of top 5% of firms by market share

	1976	1985	1996	2001	2010	2011	2012
Food and food products	65.29	70.12	75.16	65.93	75.63	73.51	79.72
Beverages	55.64	62.68	74.26	76.27	92.46	91.57	93.14
Textiles	52.29	55.92	48.11	36.00	60.77	60.26	62.79
Clothing, except footwear	46.75	50.58	58.68	34.18	68.47	68.22	73.89
Leather and products from leather	37.17	50.25	67.86	27.69	75.34	78.00	78.17
Footwear	36.73	46.08	56.42	39.99	54.56	55.48	54.10
Wood and wood and cork products	51.35	63.34	61.10	38.45	63.08	70.35	65.32
Furniture	53.39	52.12	58.38	56.68	62.28	63.98	64.69
Paper and paper products	53.36	75.43	62.05	78.13	85.55	85.22	85.17
Printing, publishing and allied industries	60.99	62.45	69.25	48.90	71.28	70.45	73.46
Basic chemicals	69.55	62.88	70.79	68.55	75.66	78.80	86.04
Other chemicals	71.32	47.99	63.43		82.76	83.08	78.15
Rubber products	55.97	66.16	80.85	40.33	77.44	75.70	72.46
Plastic products	36.55	46.63	56.67	30.22	79.39	81.25	61.48
Glass and glass products	53.46	85.40	87.31	69.74	61.99	77.32	76.79
Other non-metals	69.60	75.83	74.96	60.07	71.52	73.11	70.44
Basic iron and steel industries	73.48	76.93	69.89	76.00	83.26	83.67	82.49
Non-ferrous metal basic industries	47.60	63.07	64.66	70.60	88.45	89.23	87.55
Metal products, except machinery and equipment	58.48	65.47	67.34	47.49	60.52	58.46	60.13
Machinery, except electrical	56.14	60.24	61.79	38.41	69.86	75.01	82.47
Electrical machinery apparatus	60.77	66.58	58.26	51.60	78.76	77.84	75.36
Motor vehicles, parts and accessories	79.42	83.90	85.19	78.87	84.01	84.97	87.19
Transport equipment	68.01	73.37	75.27	58.99	70.60	76.29	75.97
Other manufacturing industries	53.15	59.90	83.38	50.66	61.88	76.60	79.44

Source: Authors' calculations.
3-Digit Industrial Classification as in Fedderke and Szalontai (2009) used.
1976–1996 data taken from Fedderke and Szalontai (2009). 2001 data taken from Fedderke and Naumann (2011).

CR_4

Table 8. Concentration ratio of top 4 by market share

	2010	2011	2012
Food and food products	30.01	30.53	38.42
Beverages	58.22	58.13	54.17
Tobacco	90.12	65.04	80.85
Textiles	20.86	19.16	18.05
Clothing, except footwear	28.96	27.08	27.87
Leather and products from leather	42.75	46.18	41.45
Footwear	33.89	33.80	28.53
Wood and wood and cork products	31.78	39.52	36.01
Furniture	35.94	38.96	36.93
Paper and paper products	59.89	56.41	56.85
Printing, publishing and allied industries	22.07	22.98	29.13
Coal and refined petroleum	81.68	82.22	86.15
Basic chemicals	28.77	29.34	55.22
Other chemicals	41.11	42.58	34.08
Rubber products	53.39	52.45	52.20
Plastic products	58.18	59.16	23.64
Glass and glass products	41.17	64.11	61.16
Other non-metals	34.01	34.22	38.27
Basic iron and steel industries	54.68	54.15	53.33
Non-ferrous metal basic industries	62.64	60.64	64.52
Metal products, except machinery and equipment	15.74	11.96	12.89
Machinery, except electrical	18.57	23.22	42.54
Electrical machinery apparatus	19.97	18.88	19.07
Television, radio and communications equipment	23.20	25.36	37.22
Professional and scientific equipment	16.26	16.65	14.54
Motor vehicles, parts and accessories	29.03	28.21	27.25
Transport equipment	31.90	38.49	40.93
Other manufacturing industries	15.83	45.47	50.30

Source: Authors' calculations.

Seventh Edition Standard Industrial Classification Used.

Table 9. HHI of top 50 firms by market share

	2010	2011	2012
Food and food products	388	400	643
Beverages	1,345	1,551	1,248
Tobacco	4,984	1,393	3,647
Textiles	209	167	147
Clothing, except footwear	310	276	272
Leather and products from leather	639	736	620
Footwear	380	392	334
Wood and cork products	382	568	550
Furniture	706	792	632
Paper and paper products	1,199	1,141	1,149
Printing, publishing and allied industries	206	218	331
Coal and refined petroleum	2,038	2,122	1,935
Basic chemicals	319	334	1,860
Other chemicals	708	765	435
Rubber products	830	823	767
Plastic products	2,467	2,460	201
Glass and glass products	1,269	1,943	1,739
Other non-metals	481	495	545
Basic iron and steel industries	1,360	1,254	1,031
Non-ferrous metal basic industries	1,317	1,170	1,286
Metal products, except machinery and equipment	88	62	73
Machinery, except electrical	156	225	828
Electrical machinery apparatus	169	157	144
Television, radio and communications equipment	225	247	397
Professional and scientific equipment	149	146	130
Motor vehicles, parts and accessories	290	295	294
Transport equipment	416	505	703
Other manufacturing industries	120	1,615	2,070

Source: Authors' calculations.
Seventh Edition Standard Industrial Classification Used.

Cartel Arrangements

- Conditions that influence the formation of cartels
  - most common in oligopoly market structures
  - small number of large firms in the industry
  - geographical proximity of the firms
  - homogeneous products that do not allow differentiation
  - stage of the business cycle
  - difficult entry into industry
  - uniform cost conditions, usually defined by product homogeneity

There are five banks competing in a local market. Each of the five banks have a 20 percent market share.

CR_4

What is the four-firm concentration ratio?

$$C_4 = 0.2 + 0.2 + 0.2 + 0.2 = 0.8$$

What is the HHI?

$$HHI = 10{,}000\left((.2)^2 + (.2)^2 + (.2)^2 + (.2)^2 + (.2)^2\right) = 2{,}000$$

Limitations

- Concentration ratios tell us nothing about the competitive structure of the rest of the industry
  - Are all the firms relatively large or are they small?
  - When the remaining firms are large, they are not as easily dominated by the top four as are dozens of relatively small firms

- Global markets
  - Foreign producers excluded

- National, Regional and local markets
  - How to factor in location?

- Industry definitions and product classes
  - 3 digit, 6 digit, etc

Rothschild Index

- Measure of elasticity of industry demand for a product relative to that of an individual firm
- How price sensitive an indiv firm's demand is relative to the entire market

$$R = \frac{E_T}{E_F}$$ total / indiv.

  - E_T = elasticity of demand for the total market
  - E_F = elasticity of demand for the product of an individual firm.
  - *R* has a value between 0 (perfect competition) and 1 (monopoly).
- Index takes value between 0 and 1
- 1: Individual firm faces demand curve that has same price sensitivity to price as the market demand curve
- $E_F > E_T$: Ind firms Qd is more sensitive to a price increase than the industry as a whole

Rothschild Index

TABLE 7–3 Market and Representative Firm Demand Elasticities and Corresponding Rothschild Indexes for Selected U.S. Industries

Industry	Own Price Elasticity of Market Demand	Own Price Elasticity of Demand for Representative Firm's Product	Rothschild Index
Food	−1.0	−3.8	0.26
Tobacco	−1.3	−1.3	1.00
Textiles	−1.5	−4.7	0.32
Apparel	−1.1	−4.1	0.27
Paper	−1.5	−1.7	0.88
Printing and publishing	−1.8	−3.2	0.56
Chemicals	−1.5	−1.5	1.00
Petroleum	−1.5	−1.7	0.88
Rubber	−1.8	−2.3	0.78
Leather	−1.2	−2.3	0.52

Source: Matthew D. Shapiro, "Measuring Market Power in U.S. Industry," National Bureau of Economic Research, Working Paper No. 2212, 1987.

Own-Price Elasticities of Demand and Rothschild Indices

Industry	*Elasticity of Market Demand*	*Elasticity of Firm's Demand*	*Rothschild Index*
Food	-1.0	-3.8	0.26
Tobacco	-1.3	-1.3	1.00
Textiles	-1.5	-4.7	0.32
Apparel	-1.1	-4.1	0.27
Paper	-1.5	-1.7	0.88
Chemicals	-1.5	-1.5	1.00
Rubber	-1.8	-2.3	0.78

Some Real-World Price Elasticities of Demand

Product	Estimated Elasticity	Product	Estimated Elasticity
Books (Barnes & Noble)	–4.00	Bread	–0.40
Books (Amazon)	–0.60	Water (residential use)	–0.38
DVDs (Amazon)	–3.10	Chicken	–0.37
Post Raisin Bran	–2.50	Cocaine	–0.28
Automobiles	–1.95	Cigarettes	–0.25
Tide (liquid detergent)	–3.92	Beer	–0.29
Coca-Cola	–1.22	Catholic school attendance	–0.19
Grapes	–1.18	Residential natural gas	–0.09
Restaurant meals	–0.67	Gasoline	–0.06
Health insurance (low-income households)	–0.65	Milk	–0.04
		Sugar	–0.04

- **Table 7.5**

•Estimated real-world price elasticities of demand

Price Elasticity of Demand for Breakfast Cereal

- What is the price elasticity of demand for breakfast cereal?
- The answer depends on whether you mean:
  - A particular brand of a particular breakfast cereal
  - A particular category of breakfast cereal
  - Breakfast cereal in general

- The further down the list we go, the more broadly the market is defined, and hence the fewer close substitutes are available.
  - So we would expect the price elasticity of demand to become smaller as we move down the list.
  - And so it does:

Cereal	Price elasticity of demand
Post Raisin Bran	–2.5
All family breakfast cereals	–1.8
All types of breakfast cereal	–0.9

Pricing Behavior

- The Lerner Index

$$L = \frac{P - MC}{P}$$

  - A measure of the difference between price and marginal cost.
  - An index from 0 to 1.

    $L = 0$
    - When P = MC, the Lerner Index is zero; the firm has no market power.
    - A Lerner Index closer to 1 indicates relatively weak price competition; the firm has market power.

- Markup Factor
  - Rearranging the above formula,

$$P = \left(\frac{1}{1-L}\right)MC$$

  - 1/(1-L) is the markup factor.
    - When the Lerner Index is zero (L = 0), the markup factor is 1 and P = MC.
    - When the Lerner Index is 0.20 (L = 0.20), the markup factor is 1.25 and the firm charges a price that is 1.25 times marginal cost.

Lerner Indices & Markup Factors

Industry	*Lerner Index* L	*Markup Factor* $\frac{1}{1-L}$
Food	0.26	1.35
Tobacco	0.76 a lot of market power	4.17
Textiles	0.21 minimal market power	1.27
Apparel	0.24	1.32
Paper	0.58	2.38
Chemicals	0.67	3.03
Petroleum	0.59	2.44

dynamics very diff

e.g price is 1,35 higher than MC for food

Lerner Indices and Markup Factors

TABLE 7–5 Lerner Indexes and Markup Factors for Selected U.S. Industries

Industry	Lerner Index	Markup Factor
Food	0.26	1.35
Tobacco	0.76	4.17
Textiles	0.21	1.27
Apparel	0.24	1.32
Paper	0.58	2.38
Printing and publishing	0.31	1.45
Chemicals	0.67	3.03
Petroleum	0.59	2.44
Rubber	0.43	1.75
Leather	0.43	1.75

Source: Michael R. Baye and Jae-Woo Lee, "Ranking Industries by Performance: A Synthesis," Texas A&M University, Working Paper No. 90-20, March 1990; Matthew D. Shapiro, "Measuring Market Power in U.S. Industry," National Bureau of Economic Research, Working Paper No. 2212, 1987.

TABLE 10.2 The Optimal Markup

another way to figure out if firm is marking up optimally

TABLE 10.2 The Optimal Markup

$$m = \frac{-1}{(1 + e_p)}$$

price elasticity of demand

ELASTICITY (e_p)	CALCULATION	MARKUP (PERCENT)
–2.0	$m = -[1/(1-2)] = +1.00$	1.00 or 100
–5.0	$m = -[1/(1-5)] = +0.25$	0.25 or 25
–11.0	$m = -[1/(1-11)] = +0.10$	0.10 or 10
∞	$m = -[1/(1-\infty] = 0$	0.00 (no markup)

Integration and Merger Activity

- Vertical Integration
  - Where various stages in the production of a single product are carried out by one firm.
- Horizontal Integration
  - The merging of the production of similar products into a single firm.
- Conglomerate Mergers
  - The integration of different product lines into a single firm.

DOJ/FTC Merger Guidelines

- Based on HHI = $10{,}000 \ S \ w_i^2$
- Merger may be challenged if
    - HHI exceeds 2500 (old 1800), or would be after merger, and
    - Merger increases the HHI by more than 200 (old 100)
- But…
    - Recognizes efficiencies: "The primary benefit of mergers to the economy is their efficiency potential…which can result in lower prices to consumers…In the majority of cases the *Guidelines* will allow firms to achieve efficiencies through mergers without interference…"

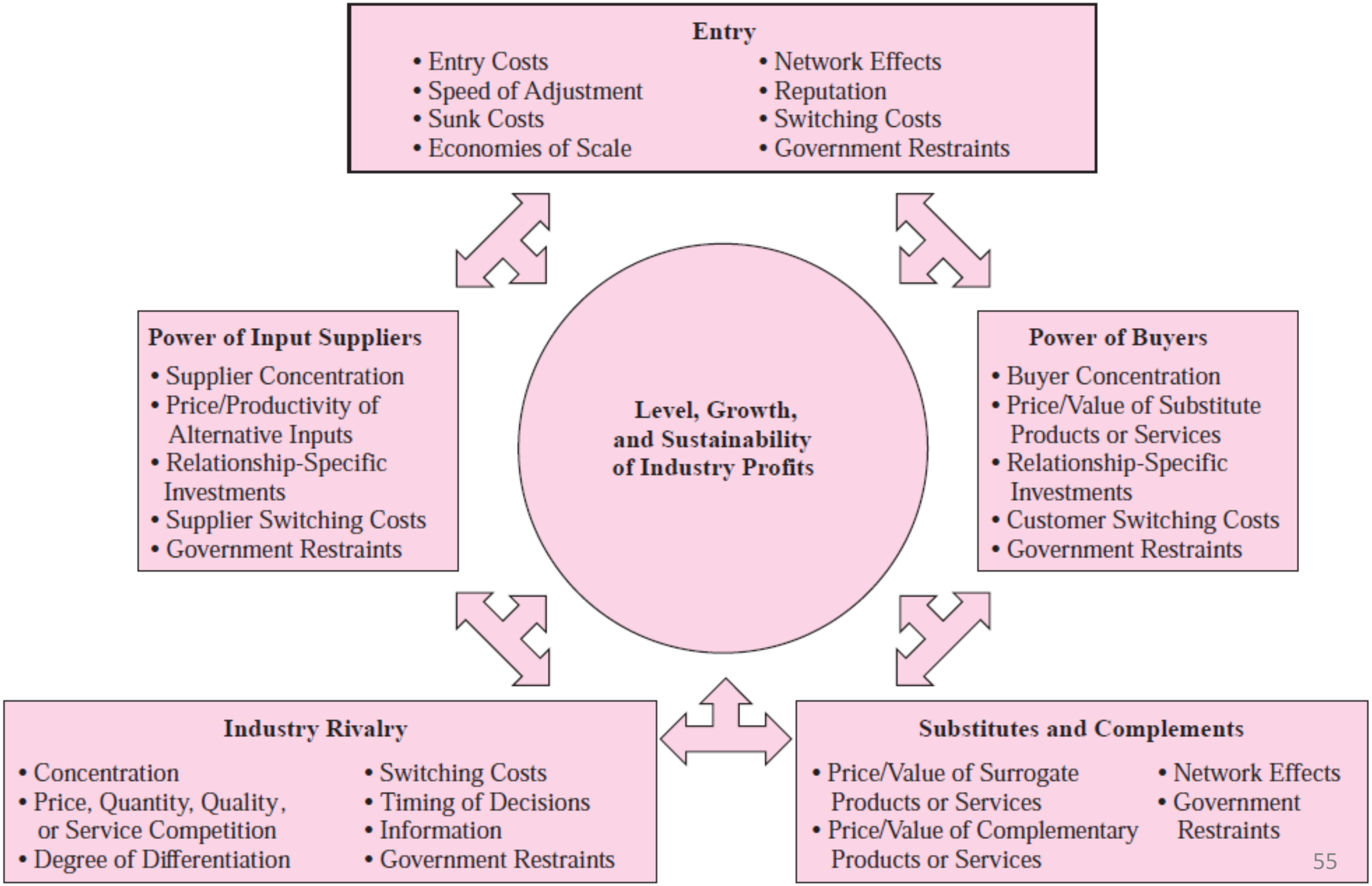
Performance

- Performance refers to the profits and social welfare that result in a given industry
- Social Welfare = CS + PS
- Dansby-Willig Performance Index
    - Ranks industries according to how much *social welfare* would improve if firms within each industry expanded output in the socially efficient manner.

Dansby-Willig Performance Index

Industry	*Dansby-Willig Index*
Food	0.51
Textiles	0.38
Apparel	0.47
Paper	0.63
Chemicals	0.67
Petroleum	0.63
Rubber	0.49

The Five Forces Framework with Feedback Effects



Pricing Strategy

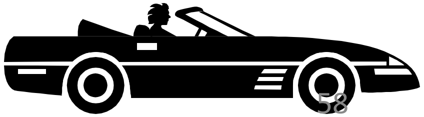
- Competitive Advantage and value creation
- Marketing position, segmentation and targeting
- Intertemporal Pricing
- Price Discrimination
- Pricing in Practice

Nature of competitive advantage

- Market conditions
  - External factors
  - Five forces analysis
  - Uncontrollable factors affecting demand (incl costs)
    - Explains 19% of variation in firms profits
- Competitive Advantage
  - Internal factors
    - A firm's ability to create more value than its competitors
      - Explains 32% of variation in firms profits

Proactive *Value-Based* Pricing

- If the price doesn't fit what customers are willing to pay, then the product may ***not*** be profitable.
- ***Customer value*** is the focus for pricing, ***not*** just the costs associated with the product.
- **Apple Computer** lost market share by ignoring customer value.
- The **Ford Mustang** was a success, as Ford found that people wanted a sports car, but didn't want it to be too expensive. The started with a price and designed the product.
    - The Mustang used value-based, ***not*** cost-plus pricing



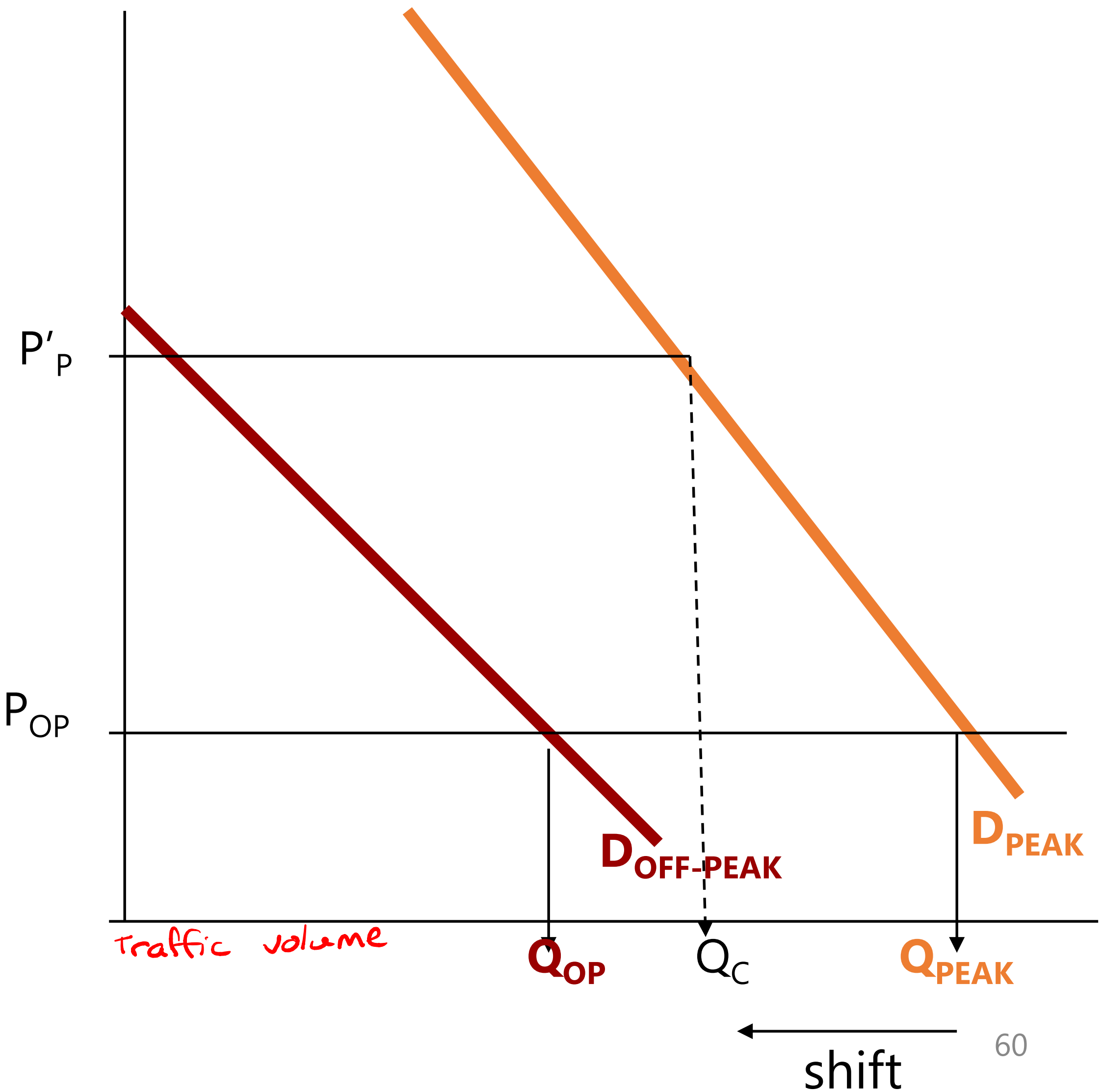
Intertemporal Pricing

- If at ***peak rush hour***, the toll is higher than at the off-peak, we are using different prices at different time periods.
- The peak toll can encourage shifting travel patterns to ***off-peak times*** or discourage some commuting altogether.
- **Intertemporal pricing** appears more frequently than one thinks. This is just one variety of what is called price discrimination.

END OF L5

Congestion Tolls

- If the price at off-peak is P_{OP} is the same price as the peak, the traffic volume varies from Q_{OP} to Q_{PEAK}.
- If the price at the peak is P'_P, the traffic volume varies less, from Q_{OP} to Q_C.



Price Discrimination

- ***Price Discrimination*** — Goods which are **NOT** priced in proportion to their marginal cost, even though technically similar
- Some Necessary Conditions:

1. **Some Monopoly Power**
   - Otherwise, in pure competition, P = MC
2. **Limited Ability to Arbitrage**
   - Separate customers and prevent reselling

Arbitage – Buy Low to Sell Higher

- Arbitrage of **Goods** is **Easy**
  - Price discrimination of goods is ***not*** effective
  - Little price discrimination of ***grocery items***
- Arbitrage of **Services** is **Difficult**
  - Price discrimination of services is effective
  - Price discrimination at ***restaurants by age***, as restaurant food is a service
  - Lawyers charge different prices for wills, based on ability to pay

TABLE 10.7 Types of Price Discrimination

TABLE 10.7 Types of Price Discrimination

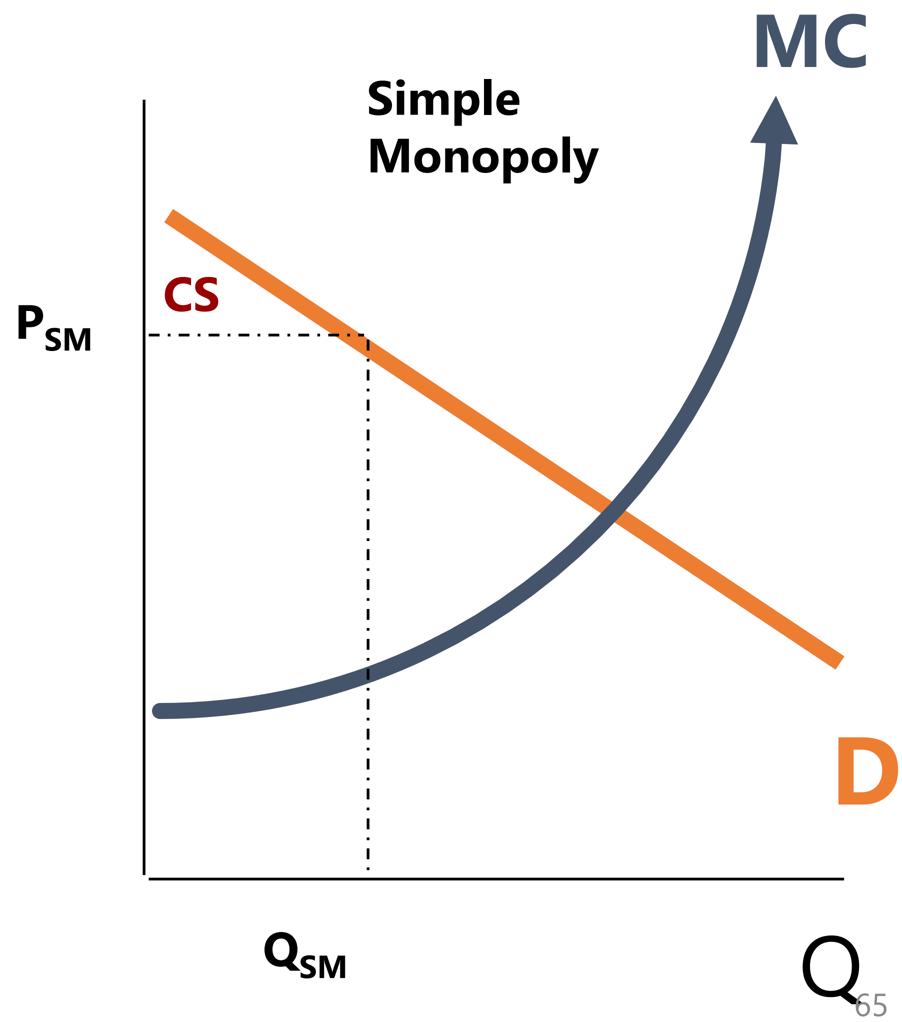
TYPE	DESCRIPTION
First-degree or personalized pricing zero CS	Charging each individual the maximum amount he or she is willing to pay for each unit of the product.
Second-degree	Charging the maximum price consumers are willing to pay for different blocks of output.
Third-degree or group pricing	Separating the markets into different groups of consumers and charging a higher price (relative to cost) in the market with the more inelastic demand. Sometimes called zone pricing.
Versioning	Offering different versions of a product to different groups of customers at various prices, each designed to meet the needs of the specific groups.
Bundling	Selling products separately, but also as a bundle, where the price of the bundle is less than the sum of the prices of the individual products.
Promotional pricing	Using coupons and sales to lower the price to customers with more elastic demands, but also imposing costs on these customers.
Two-part pricing	Charging consumers a fixed fee for the right to purchase a product and then a variable fee that is a function of the number of units purchased.

Ways to Separate Customers for Price Discrimination

1. **Geography** as when the price in the East-side and West-side differ
2. **Income** as the Canadian Econ Association charges more to professors than students
3. **Gender** as when jeans for women are priced higher than similar jeans for men
4. **Age** as when kids get in at lower prices for movies
5. **Time** of day or season
6. **Race** as when shampoos targeted for Afro-Canadian hair are priced differently that other shampoos, though technically the same.
7. **Language** as when products printed in Spanish are priced differently than those in English/French
8. **Transient/Resident** as when contractors pay less at hardware stores than other customers
9. **Ability to Haggle** when those who ask for a lower price get it

Why Price Discriminate?

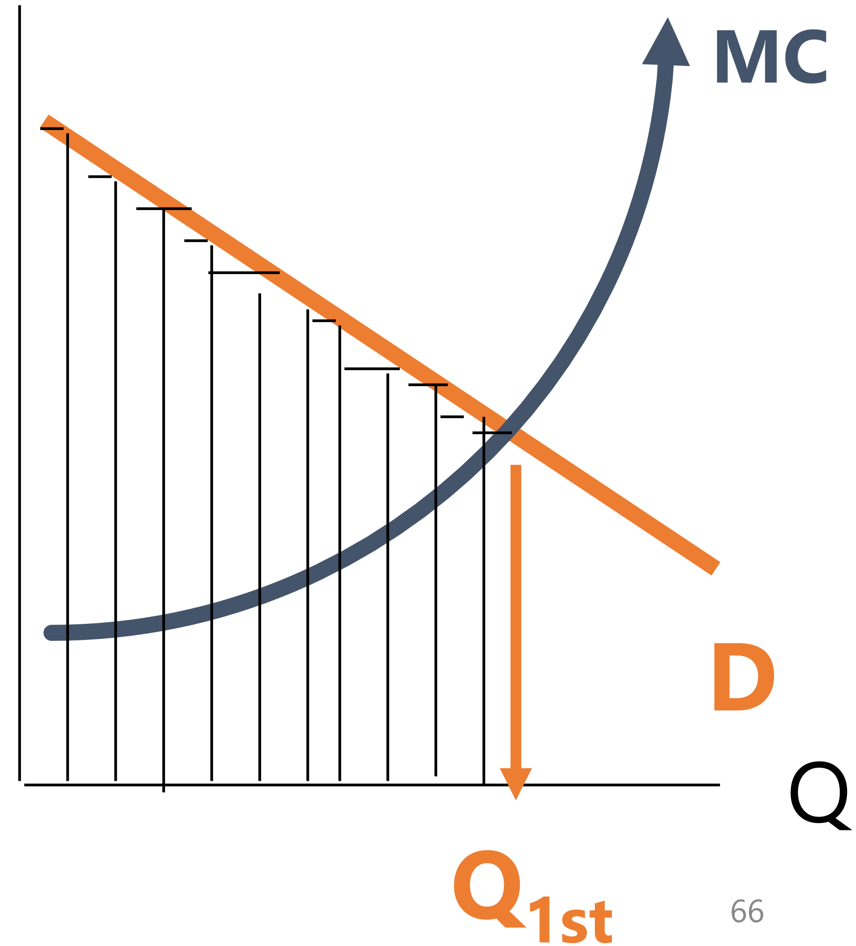
- In **Simple Monopoly**, there is only one price
- Consumers receive a **consumer surplus**
- In Price Discrimination, monopolists can **SCOOP OUT all** consumer surplus



Perfect Price Discrimination
(or 1st Degree Price Discrimination)

- Charge the **MOST** that a person is willing to pay for each good
- Zero consumer surplus
- Produce **MORE** than in Simple Monopoly
- Output the same as in Competition

Price Discriminating Monopoly



Notice: Incentives to Understate One's True Willingness to Pay

- The conditions for perfect price discrimination are seldom met
- Hence, some close approximations exist

Second Degree Price Discrimination:
Units are Grouped

- There are are a variety of ways to group units to attempt to scoop out consumer surplus

Second Degree Price Discrimination: Two Part-Pricing

- A price for the privilege of buying items PLUS a price per item
- Examples:
  - Car rental per day with per kilometre charges
    - Car renters may not know how much they will use the car.
    - They may prefer a lower rental rate (cover charge) with a per kilometre charge.
  - Amusement parks
  - Country Club Dues and Greens Fees
  - Cover Charge to Enter a Bar and a Price Per Drink

Second Degree Price Discrimination at McDonalds (Bundling)

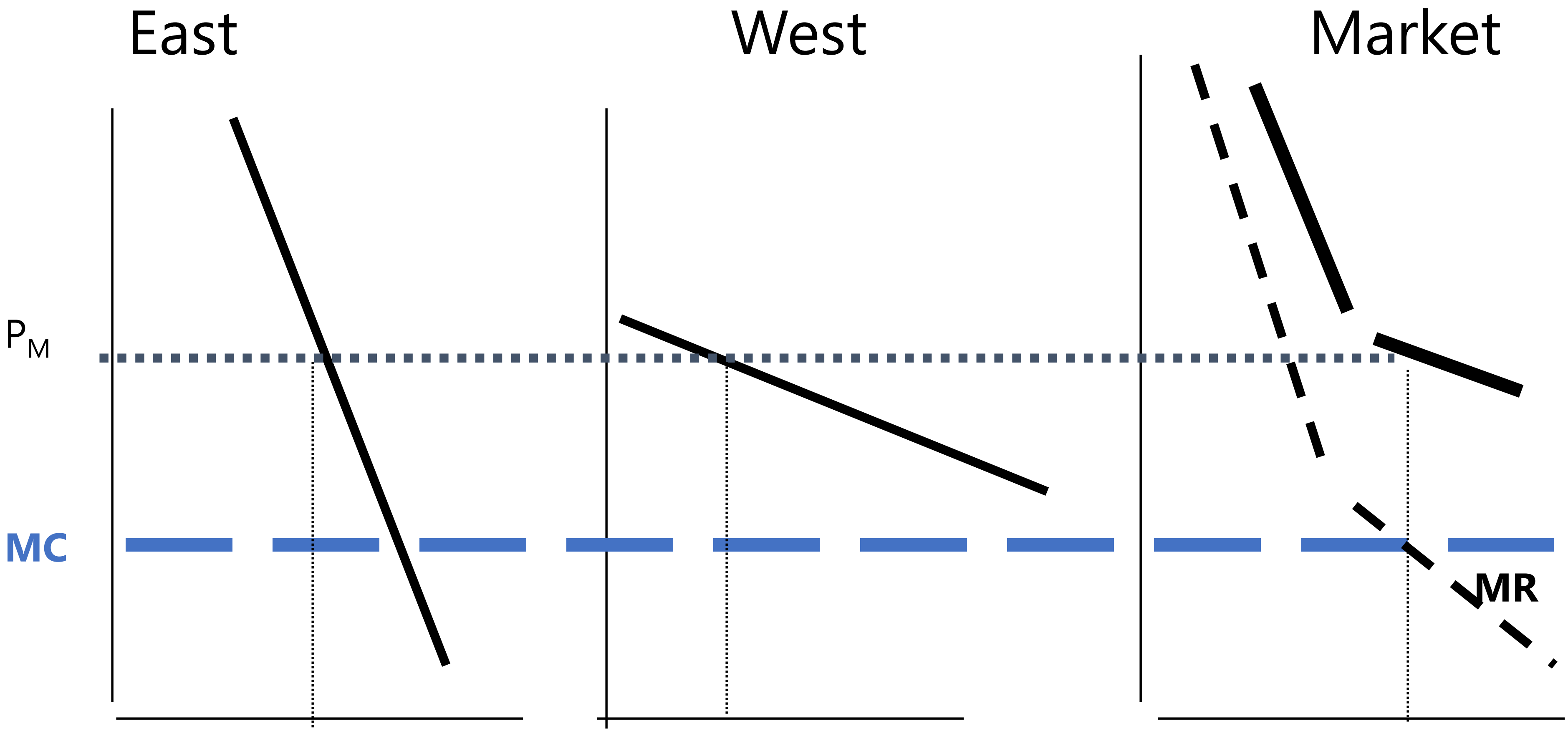
- McDonalds sells *Extra Value Meals*, as a bundle of sandwich, fries, and a soft drink for ***less*** than it sells them separately.

- Selling both bundles and items separately is **mixed_bundling**.

Bundling & Mixed Bundling

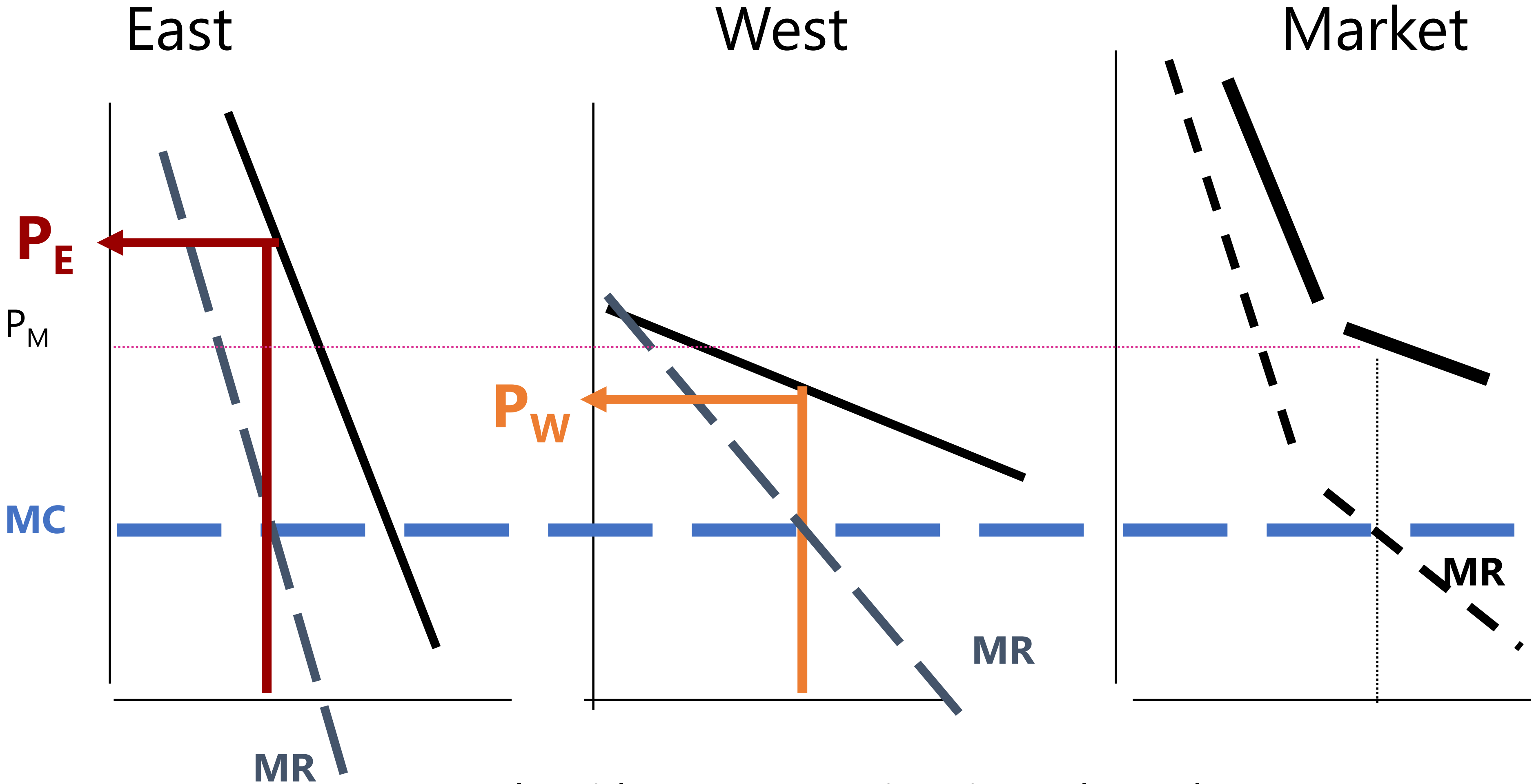
- If Bob would pay $3 for a burger and $1 for a soft drink, and if Mary would pay $2 for a burger and $2 for a soft drink, a bundle of $4 for both a burger and soda will work for both customers as a bundle.
- But if the price of a burger individually were $3.00 and a soft drink $2.00, then Bob would buy only a burger and Mary only a soft drink.
    - Not everyone is alike, so mixed bundles succeeds with more customers.

One Price for All Regions



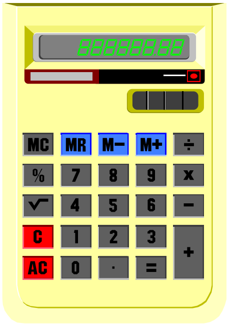
Example with a **Simple Monopoly Price** (P_M) in both markets

Third Degree Price Discrimination



Example with **Different** Prices in Each Market

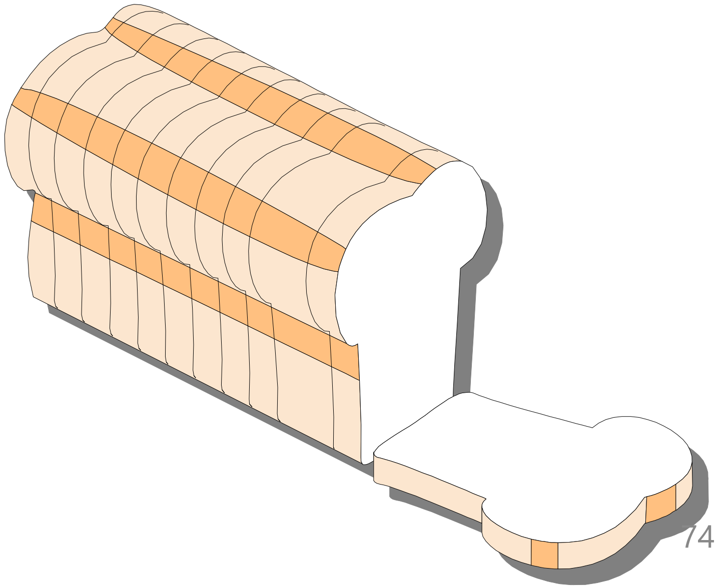
Pricing in Practice



- In practice, pricing strategy involves the whole **life-cycle pricing** of the product.
- Managers report wide use of **cost-plus pricing** methods because it:
  - Streamlines pricing of multiple products
  - Streamlines pricing of retail products

Optimal Markups in Practice

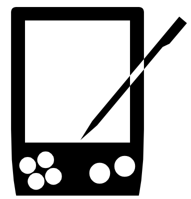
- **Grocery stores** have low markups
- Many close substitutes -- at other grocery stores (bread varieties and qualities are standardized)
- Frequent purchase, so customers are knowledgeable about prices & quality
- Demand is therefore **highly elastic**
- Optimal markup would consequently be small



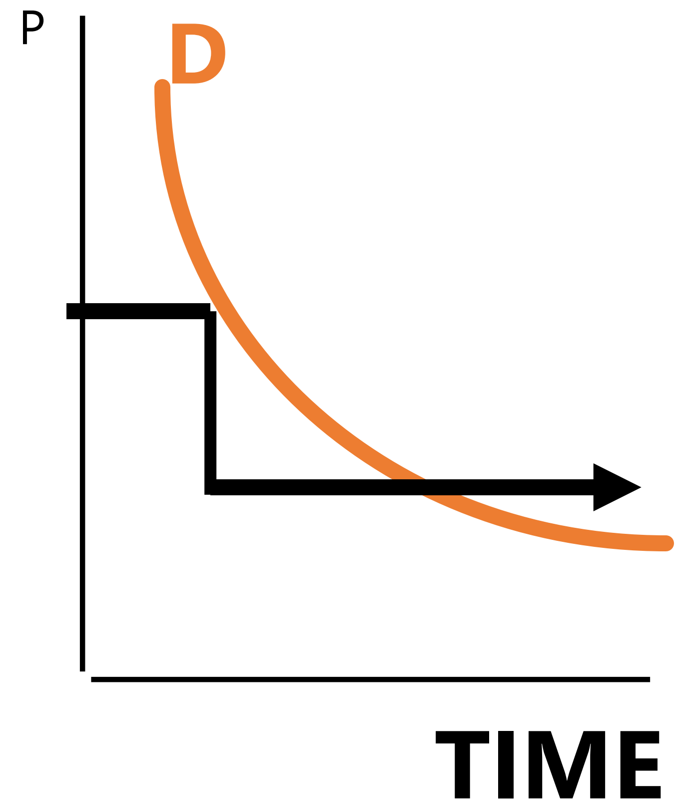
Markups on Jewellery

- Jewellery markups are known to be large
- Difficult to make comparisons across jewellery stores
- Little repeat purchases, so knowledge about prices is low
- Consequently, lower price elasticity for jewellery
- The optimal markup is larger

Skimming



- Price declines over time
- Those who wish to get it first pay the highest price, others are willing to wait
- ***Examples:***
  - Hardcover & Paperback Books
  - New electronic, computer products, and PDAs.



Prestige Pricing

- Some products distinguish themselves by being noticeably **expensive**.
  - Mercedes, Rolls Royce, or BMW
  - Cartier jewellery
- Price is a way to distinguish the product
- **Prestige Pricing** is the practice of charging a high price to enhance its perceived value.
  - However, firms spend much on promotional activities to convince customers that the product is prestigious.

Other Pricing Practices

- Price skimming
  - the first firm to introduce a product may have a temporary monopoly and may be able to charge high prices and obtain high profits until competition enters
- Penetration pricing
  - selling at a low price in order to obtain market share

Other Pricing Practices

- Limit pricing
  - A monopolist will set price below MR = MC to prevent potential customers from entering the market.
- Predatory pricing
  - Setting price below marginal cost to drive competitors out of the market

Other Pricing Practices

- Prestige pricing
  - demand for a product may be higher at a higher price because of the prestige that ownership bestows on the owner
- Psychological pricing
  - demand for a product may be quite inelastic over a certain range but will become rather elastic at one specific higher or lower price